a flood of information that needs to be filtered and communicated to the driver. Moreover, the driver expects the vehicle to be a digital companion. As a result, solutions are required that ensure intuitive, easy, and, above all, safe interaction between the driver and vehicle.

The smart voice assistant is an intelligent speech assistant that can learn and adapt to user behaviour, preferences, and context. It provides fluent and intuitive natural language interaction with context awareness and seamless transition between various functions. For example, the driver might ask for route planning for navigation. The driver might then ask about free parking spaces at the destination and even dictate a text message to book a table at a restaurant in the vicinity of the destination. The system ensures coherent communication, sends the relevant data from the navigation system to the parking space assistant, matches the Internet search for restaurants to the location of the recommended parking garage, and, finally, transfers data from the navigation system (estimated time of arrival) and restaurant search (address and e-mail contact) to the e-mail and voice recorder program in order to reserve a table.

The intelligent assistant even understands the request "search for restaurants there" and correctly interprets "there" as the previously selected destination. The intelligent assistant also detects meaningful connections without the driver having to issue standard commands now and then. The assistant can also handle multiple questions, or two tasks communicated in a single sentence. If the driver says, "I would like to get to the mall as quickly as possible and eat Chinese food somewhere nearby," for example, the assistant will calculate the route and also search for Chinese restaurants near the destination.

As we move towards autonomous and connected driving, the amount of information to be displayed on the screens also increases. The conventional 2D display would no longer be sufficient. However, the usual 3D displays require additional gear that hampers the in-vehicle experience for the driver and passengers. The new Natural 3D Lightfield display is an evolutionary

The vehicle becomes an always helpful and smart companion

step in the design of human-machine interaction of vehicles and brings out a new measure of safety and comfort by generating an unprecedented 3D experience without special eyewear or head-tracker cameras.

With this innovative technology, information can be safely conferred to the driver in real time, which allows the interaction between the driver and the vehicle to become more intuitive and comfortable. The safety is enhanced with Natural 3D lightfield as it provides a superior usability, because the perception of information can be accelerated as this visual experience is closer to the real world.

This 3D experience is not limited to driver as the passenger can also experience this in the pillar to pillar displays. The 3D image produced by the Lightfield display consists of a total of eight perspectives of the same object that vary subtly according to the point-of-view. The display uses parallax barriers, slanted slats that divide the image for the viewer. As if looking at real objects, two different, slightly offset views reach the right and left eye, resulting in a three-dimensional image.

Not just the display, the audio systems are also transforming. The speakerless audio system that uses Ac2ated Sound, enhances the vehicle's interior with its immersive sound quality. The technology replaces conventional loudspeaker technology with actuators that create sound by vibrating certain surfaces in the vehicle to produce sound. The speakerless audio system is a revolutionary system, where 3D sound reproduction surrounds the passengers in a detailed and vivid soundscape to maximize their in-car audio experience. This system not only reduces the weight and space by up to 90 percent compared to the conventional audio system but also assists in enhancing the audio quality significantly.

It could be an ideal system for electric vehicles as well, where saving space and weight are top priorities. With this system, many components become unnecessary due to vibrating surfaces in the vehicle just as speaker diaphragms. Actuators cause components, such as the A-pillar trim, door trim, roof lining, and rear shelf, to vibrate so that they emit sound in different frequency ranges.

However, these HMI functions would not be possible without the change in the vehicle architecture. The cockpit high-performance computer (HPC) integrates all cockpit domains like clusters, infotainment, camera into another powerful high-performance computer, which is the centrepiece of user experience in the vehicle. The shift in architecture to move away from numerous individual control units to a few high-performance computers is to reduce cost and complexity for vehicle manufacturers over whole vehicle lifecycle.